

AI-Based Restoration of Degraded Inspection Images

KLA Problem Statement PS01 · SemiCon India Hackathon 2026 · Round 2

Team **ForgeX**

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23.63 dB

PSNR on the organisers' test set

+3.18 dB

over bicubic ×2

0.193

LPIPS (bicubic 0.466)

1.37 M parameters · 5.5 MB · 297 images in 5.9 s end-to-end on an RTX 4050 laptop · torch + numpy only

Problem Statement Addressed

The statement asks for generalisation across distributions. Everything below was built and tested against that.

	Round 1	Round 2
Content	natural photographs	SEM micrographs
Training pairs	3,200	4,785
Sizes present	mixed	all 256→128; no 512×512
f90 — structure fineness	0.0945	0.2913 (3.1× finer)
Downsampling operator	exact 2×2 box average	exact 2×2 box average
Bicubic ×2 baseline	23.23 dB	20.46 dB

Bicubic falls 2.77 dB between rounds. Equal skill therefore shows as a lower score.

THE FOUR THINGS THE STATEMENT ASKS FOR

- Recover fine spatial detail lost to downsampling
- Suppress speckle noise across diverse image types
- Generalise broadly across multiple image distributions**
- Handle the reduction in spatial resolution (the ×2 path)

The third one is the whole deck. Every decision below was made against it.

Why f90 matters. f90 is the radial frequency below which 90% of non-DC energy sits. Round-2 structure is 3.1× finer, so 3.1× more of the signal lives above the Nyquist limit of the downsampled input — information genuinely destroyed by the 2×2 average, not merely obscured by noise. That sets a ceiling no architecture can lift, and it is why we spent the round measuring data rather than trying models.

Test & Evaluation Setup

Every number in this deck is a final-5-epoch mean, scored against a 0.005 dB measured noise floor, on a shared baseline.

THE NOISE FLOOR

0.0049 dB two runs differing only in random seed, final-5-epoch means, matched at epoch 39.
We use **0.010 dB** as the reporting threshold — deliberately conservative. Anything under it is reported as null.

FINAL-5, NEVER BEST-EPOCH

Best-epoch selection inflates a score by a **median 0.058 dB** across our six 120-epoch runs — six times the floor. Every number in this deck is a mean of the last five epochs.

ONE PASS, ONE BASELINE

All 11 models are scored by the same code on the same inputs. The shared **bicubic column is the checksum** — if it differs between two files, those files cannot be compared. We learned this the hard way.

Six ways we measure, and what each one is for

Test set	n	Distribution	What it answers
Our held-out split	200	in-distribution	day-to-day comparison between runs
Organisers' test set	297	in-distribution	the score they will see
Withheld content block	478	unseen morphology	does it work on content never trained on
Ten NFFA morphologies	240	different source	does it work on a different corpus
Nine-level noise sweep	60 × 9	ID → far OOD	does it hold when the sensor changes
Nine-axis OOD suite	40 × 9	noise / blur / kernel / content	worst-case axis, not just the mean

Zero exact overlap between the organisers' 297 images and the 4,785 training images — verified by SHA-1 over the raw arrays.

Degradation Discovery

We measured KLA’s degradation before touching a model: a three-term variance law, and speckle over shot noise by $\Delta\text{BIC} = 1118$.

$$\begin{aligned} \text{var}(\text{residual}) &= \sigma^2_{\text{add}} + \sigma^2_{\text{mul}} \cdot m^2 + c \cdot v \\ \text{downsample} &= \text{exact } 2 \times 2 \text{ box average} \end{aligned}$$

m = 2×2 block mean of the clean image, v = within-block variance. adjusted $R^2 = 0.9987$ over 288 bins / 6.55 M pixels.

Speckle or shot noise? Settled by BIC, not by assumption.

Poisson shot noise has variance linear in the mean; multiplicative speckle quadratic. They call for different variance-stabilising transforms, so we fitted four models to identical bins over 6.55 M pixels.

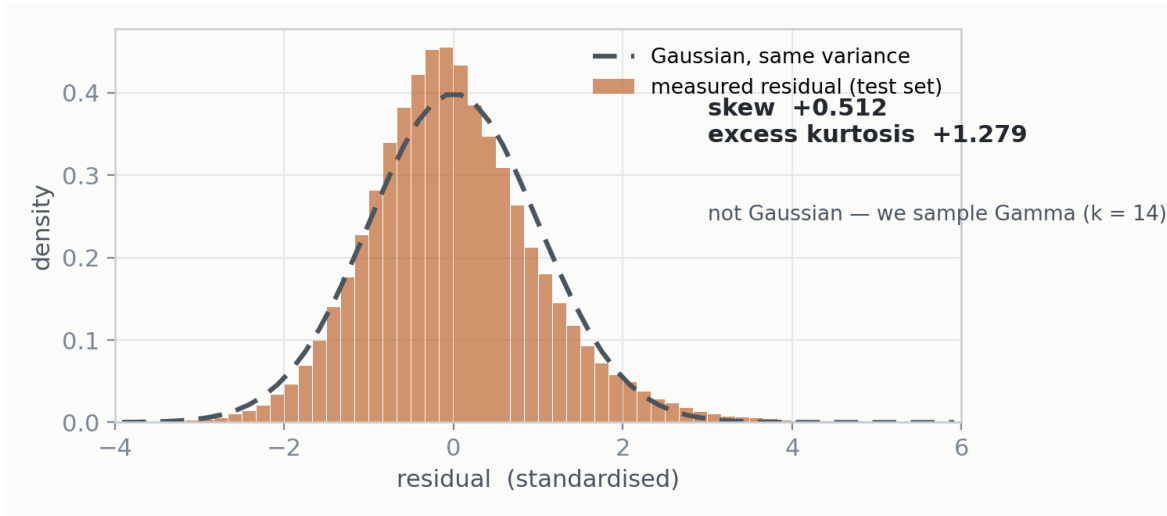
Model	form	adj R^2	BIC
M1 multiplicative	$a + b \cdot m^2 + c \cdot v$	0.9987	-1814.5
M2 Poisson	$a + p \cdot m + c \cdot v$	0.9370	-696.2
M3 both terms	$a + p \cdot m + b \cdot m^2 + c \cdot v$	0.9989	-1849.5
M4 no brightness term	$a + c \cdot v$	0.1808	+37.7

Measured on KLA’s own pairs: σ_{mul} 0.155–0.258 σ_{add} 0.000–0.065 c 0.026–0.234

$\Delta\text{BIC} = 1118.3$ — **decisive**. M3 fits marginally better but drives its Poisson coefficient negative (−0.00196), which is meaningless for shot noise. We keep M1.
Consequence: the generator multiplies rather than drawing Poisson counts. Assuming shot noise would have stabilised the wrong thing.

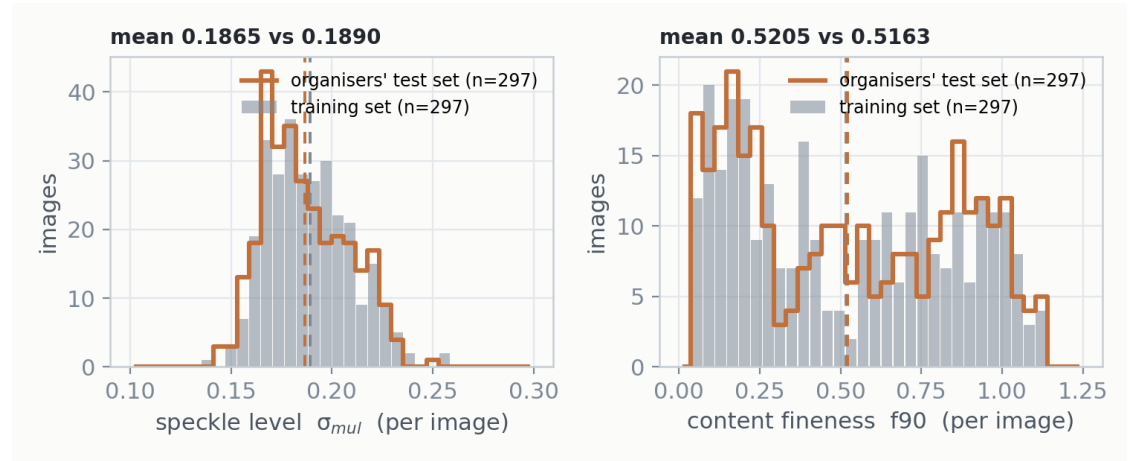
Distribution Analysis

The noise is not Gaussian, so we draw Gamma — and the organisers' test set is in-distribution, so we did not optimise for it.



WHY GAMMA, NOT GAUSSIAN

The residual is spatially white — lag-1 correlation -0.052 horizontal, -0.050 vertical — but it is not Gaussian: skew **+0.51**, excess kurtosis **+1.28**. A Gaussian generator would match the variance and miss the shape, so the network would never see the heavy right tail that real speckle produces. We sample Gamma with $k = 14$.



THE ORGANISERS' TEST SET IS IN-DISTRIBUTION

We refitted the degradation on the 297 images they shared and compared it against 297 random training images. **Every property matches** — σ_{mul} 0.1865 vs 0.1890, f90 0.5205 vs 0.5163, skew +0.512 vs +0.539, kurtosis +1.28 vs +2.08, lag-1 -0.052 vs -0.048 . Only 1–3% of test images fall outside the training p1–p99 band, which is what two samples of the *same* distribution give by construction.

The statement promises a test of generalisation across distributions. This set is not that test — so we assume it is not the final one, and we did not optimise for it.

Synthetic Degradation Generator

Two generators: one calibrated to reality, one deliberately wider — and no randomised operation order, because we measured that it was wrong.

parameter	NARROW — measured in KLA's pairs	WIDE — deliberately broader
σ_{add}	0.000 – 0.065	0.000 – 0.180
σ_{mul}	0.155 – 0.258	0.060 – 0.500
c (detail term)	0.026 – 0.234	0.000 – 0.450
Gamma k	14	4 – 60
optical blur	absent	σ 0.3–1.1 on 30% of samples
downsampling kernel	exact 2×2 box	box, or Gaussian-weighted on 25%

src/degrade.py — `degrade()` and `degrade_wide()`. 2.1 ms per call, so synthesis never starves the GPU.

ONE DESIGN DETAIL THAT MATTERS

Blur is applied to the clean image before noise is sampled — never to the output. Optics act on the image, not the sensor. Blurring the output would correlate noise we measured to be white (lag-1 -0.052), teaching the network to remove a structure that does not exist in real data.

THE MIXTURE, PRECISELY

`wide_p` is the fraction of the synthetic draws, not of all draws. The roll is sequential:
`real = p    wide = (1-p) · w    narrow = (1-p) · (1-w)`
So `--p-real 0.5 --wide-p 0.5` is **50% real / 25% wide / 25% narrow**.

CALIBRATION IS A TEST, NOT A CLAIM

We re-fit our own synthetic output with the same estimator used on the real pairs:

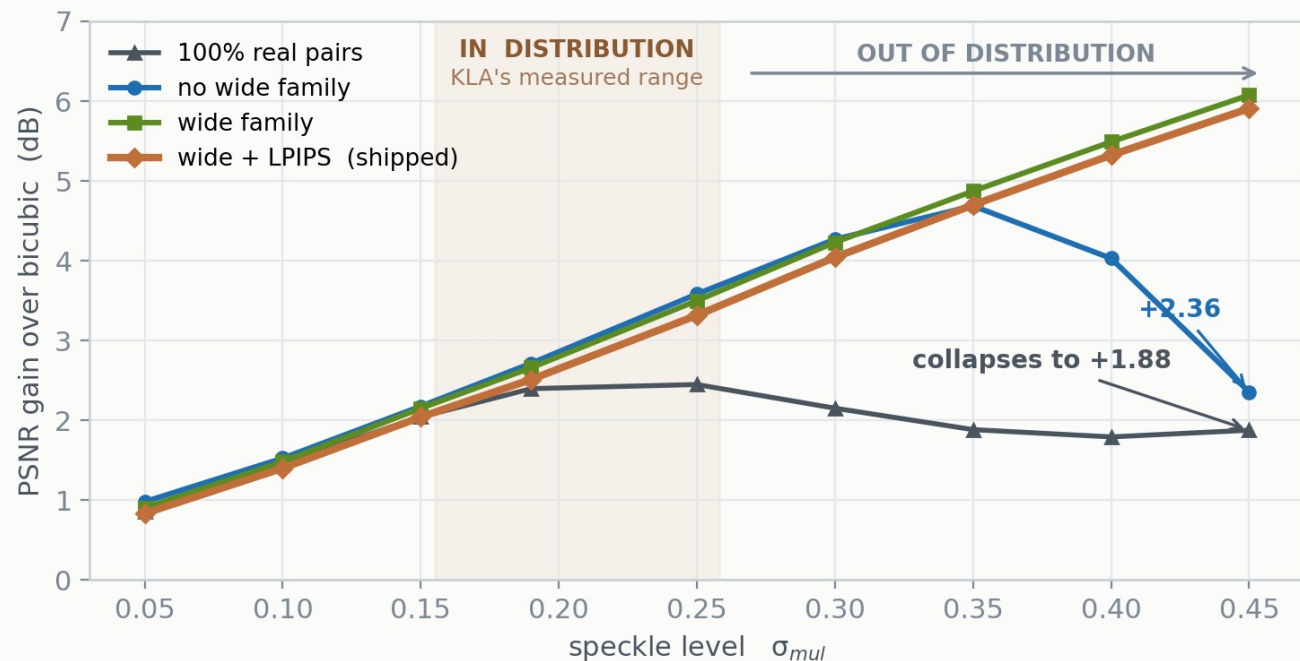
KLA real	σ 0.0414	σ 0.1863
ours NARROW	σ 0.0415	σ 0.1986
ours TRAINING	σ 0.0507	σ 0.2192

The narrow family brackets reality — that is the test. The training mixture is visibly wider on purpose.

Three changes from Round 1, each forced by a measurement: added the c·v detail term (4.4% of variance on SEM vs 1.8% on photos), removed the blur from the narrow family (residual error rose 0.01074 → 0.01235 with it), Gaussian → Gamma.

Real vs Synthetic — the Reversal

Real-only training wins in-distribution and collapses outside it. Training wider than reality costs 0.013 dB and buys 3.72 dB.



WHAT WE BELIEVED FOR A WEEK

100% real pairs beat every synthetic mixture on the held-out split — 23.267 vs 23.218 vs 23.134. Measurement sound, inference wrong. $\sigma \approx 0.19$ is where KLA's data sits. **All models agree there** — within 0.30 dB. We were resolving a 0.13 dB difference at the one operating point where there is nothing to see.

Three σ -steps out, the real-only model is 4.03 dB behind. And it does not merely stop improving — it *turns over* at $\sigma = 0.25$ and declines. A model that had learned the inverse problem would keep gaining, because there is more noise to remove. Turning over is the signature of a model that memorised a noise amplitude.

The objection we had to answer. noise_sweep.py builds its inputs with our own generator, which structurally favours a synthetic-trained model — we flagged this ourselves and nearly discarded the sweep. Two things rescue it. (1) The bias does not appear where it should: at σ 0.05–0.19 the *real-only* model beats the model trained on that very generator. (2) The conclusion reproduces on inputs the generator never touched — the ten NFFA morphologies and the withheld content block, both against real ground truth. The sweep is where we *found* the effect; it is not where we certify it.

Proposed Solution

A 1.37 M-parameter residual CNN with a bicubic skip and a measured loss — Charbonnier + SSIM + gradient + LPIPS.



Architecture

Residual CNN — EDSR-style body, all work at low resolution, PixelShuffle ×2 only in the final block. 4× cheaper than upsample-first designs, and inference time is scored.

Global bicubic skip. The network predicts only the correction to a cheap baseline, so it starts at the bicubic score and converges fast.

Variance-stabilising stem. A parameter-free layer feeds the first convolution raw x , $\sqrt{x}$ and $\log(1+x)$. It picks the representation where signal-dependent noise is closest to uniform — rather than us assuming which one that is.

No BatchNorm. Well established to hurt super-resolution; confirmed here.

64 channels · 16 residual blocks · 1,368,705 parameters · 5.5 MB

Training & loss

Charbonnier + 0.15·SSIM + 0.05·gradient + 0.05·LPIPS. No GAN.

120 epochs × 500 iterations, batch 32, 64×64 crops, AMP. Adam, cosine schedule.

The gradient term sharpens edges only where the ground truth has edges — that is the mechanism that keeps the perceptual term honest.

--seed and --split-seed are deliberately separate. The validation split must stay fixed while training randomness varies, or a seed-variance experiment compares two models on two different validation sets and measures nothing.

RTX 4050 Laptop 6 GB (75 W): 4.3 h. RTX 4090 pod: 28 min.

Peak VRAM 0.95 GB.

Results

23.63 dB on the organisers' own 297-image test set: +3.18 dB over bicubic, 59% lower perceptual distance.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Bicubic $\times 2$ (no denoising)	20.455	0.5099	0.4655
Ours — 1.37 M params ★ submitted	23.632	0.6079	0.1929
gain	+3.18 dB	+0.098	-59%

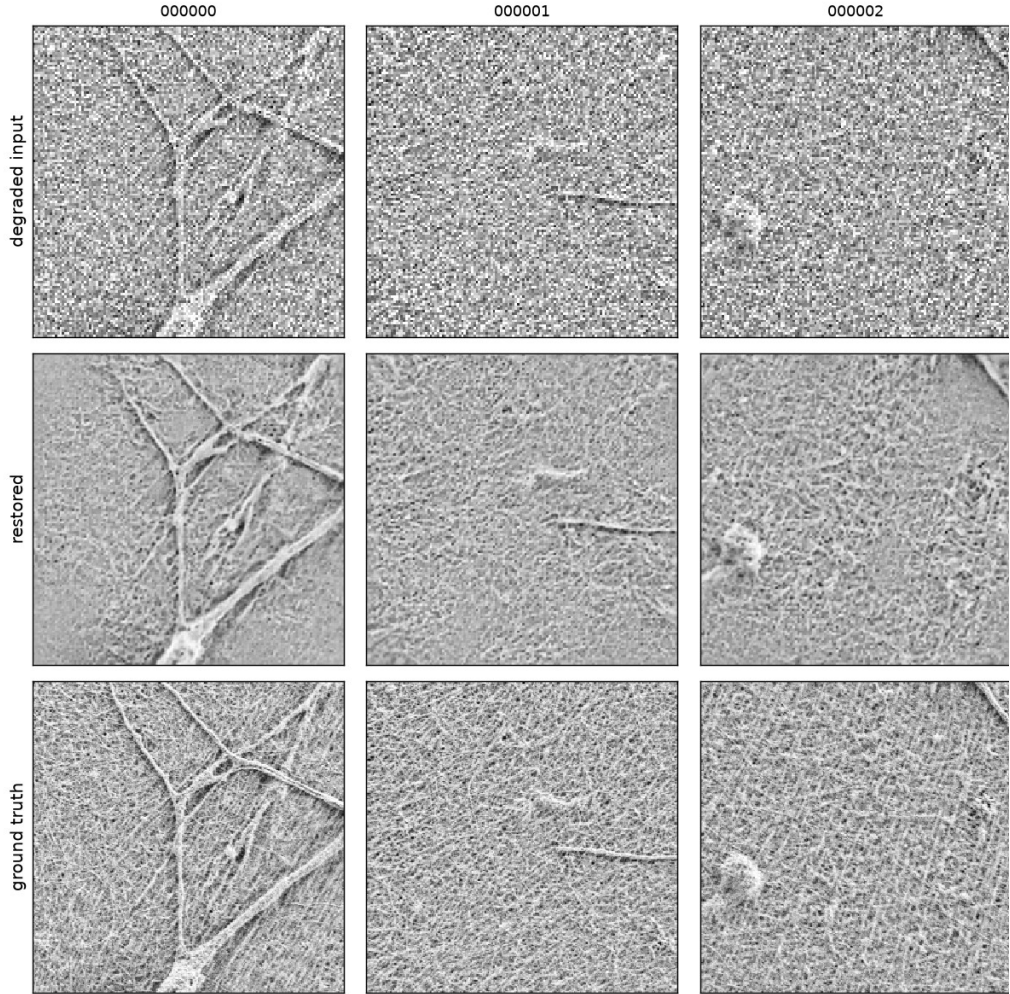
+3.18 dB over bicubic

and 59% lower perceptual distance, for 1.37 M parameters

THE SAME MODEL, EVERYWHERE ELSE WE MEASURED IT

test set	n	PSNR	vs bicubic
organisers' test set	297	23.632	+3.18
our held-out split	200	23.104	+2.88
withheld content block	478	24.302	+3.65
ten NFFA morphologies	240	—	+4.01
512×512, never trained at that scale	778	25.392	+4.52

Every row is real ground truth. The gain is largest where the model has seen least.



degraded input $\rightarrow$ restored $\rightarrow$ ground truth

Ablations

Eleven interventions measured against the noise floor. Seven did nothing. One — the wide degradation family — decided the model.

Change tested	Effect on PSNR	Verdict
Random seed alone (matched at epoch 39)	0.0049 dB	<i>the floor itself</i>
1.8× receptive field, parameters matched	+0.0077 dB	null
Degradation recalibration on its own	−0.0162 dB	null
Gaussian-weighted downsampling (soft_p)	no saved measurement	<i>undocumented</i>
Content-balanced sampling ($p \propto f90$)	−0.0903 dB	null
8× test-time augmentation	+0.0181 dB, 7.1× slower	rejected
Pinned-buffer + CUDA-stream I/O path	+0.04 s on a 1.94 s spread	rejected
channels_last layout fix	−0.06 s on a 0.73 s spread	rejected
4× parameters (343 k → 1.37 M)	+0.108 dB	<i>real, but see capacity</i>
Rebuilt generator vs Round-1 generator	+0.0836 dB	real effect
Real pairs instead of synthetic (in-distribution)	+0.0494 dB	real effect
Wide degradation family, at $\sigma = 0.45$	+3.72 dB	the effect that decided the model

Matched final-5 windows from runs/*/log.csv, or one scoring pass over identical inputs. docs/ENGINEERING_LOG.md §10.

THREE CLAIMS WE WITHDREW

“9× more training data = +0.006 dB” — the 500-image run validated on a *different* 200 images. Not comparable. Dropped.
“3× training length = +0.030 dB” — no matched pair exists in runs/. Dropped.
“2.7× parameters = +0.026 dB” — a Round-1 number. The Round-2 answer is the capacity sweep.

WHAT DOMINATES THE SCORE

Not anything we control.

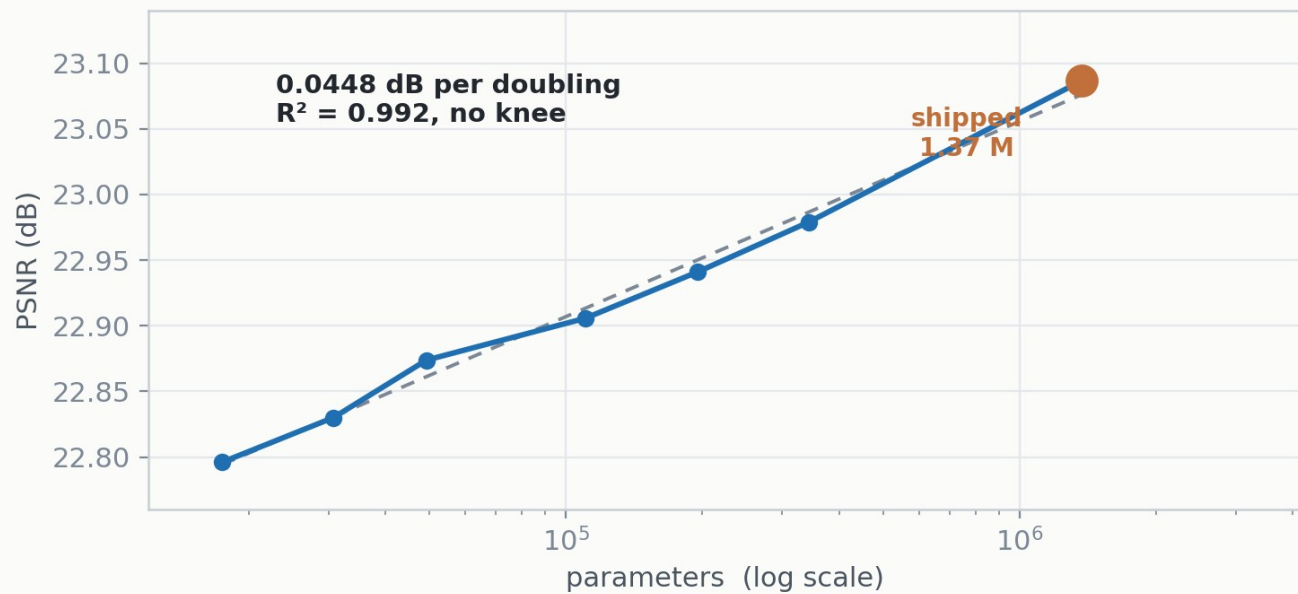
Coarsest content block 26.664 dB

Finest content block 21.274 dB

5.39 dB from content alone — 40× larger than any architectural change we measured, and 1100× the noise floor. That is why this pipeline targets robustness across content rather than peak score on one split.

Capacity & Inference

1.37 M parameters is a deliberate quality-versus-efficiency choice, not a default — and 297 images take six seconds



Seven sizes across 78.4× of parameters buy 0.291 dB in total — 0.045 dB per doubling, $R^2 = 0.992$, no knee. A capacity-starved model climbs and then flattens; this is a straight line, so the bottleneck is information the 2×2 average destroyed, not model size. A 49 k model gives up 0.21 dB and is 9× faster — we know the price of going smaller, and chose not to pay it.

END-TO-END, MEASURED THE WAY IT WILL BE SCORED

5.9 s

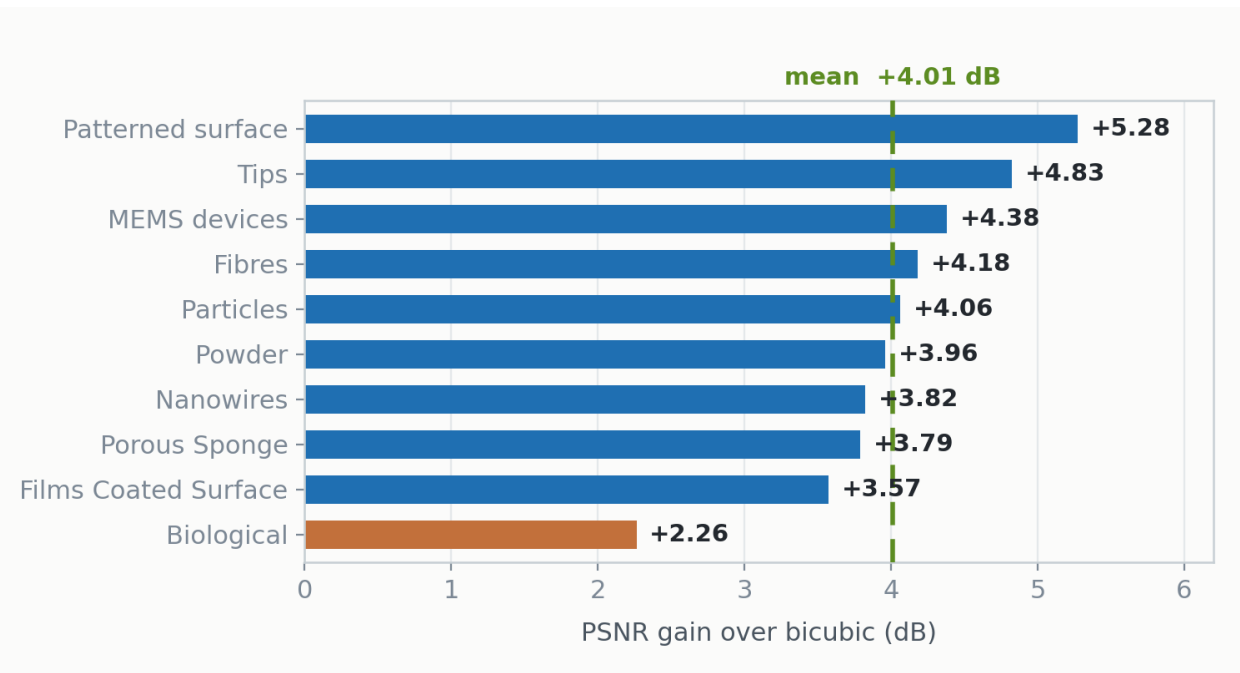
297 images, process start to last file written, RTX 4050 laptop — **19.9 ms per image**. The network itself is ~2 ms of that; roughly 72% is Python importing torch and CUDA initialising, which amortises over test-set size.

WHAT WE TRIED AND DID NOT SHIP

Three inference optimisations were written and benchmarked with alternating repetitions against the run-to-run spread: 8× test-time augmentation, a pinned-buffer + CUDA-stream I/O path, and a channels_last layout fix. **None cleared the noise; all produced bit-identical output.** ONNX export was rejected without benchmarking — it adds a dependency the submission forbids. Details in the ablation table and docs/bench_results.txt.

Generalisation

Unseen content +3.65 dB. Unseen scale +4.52 dB. Ten morphologies from a different corpus, all positive. The gain is largest where the model has seen least.



Ten labelled NFFA-EUROPE morphologies — a different corpus, and the source dataset ships the labels

All ten positive. Mean +4.01 dB — higher than on KLA's own data. LPIPS stays 0.199–0.299 across all ten. The 3.01 dB spread orders morphologies exactly as the f90 analysis predicts: regular geometric structure easiest, soft organic structure hardest. Honest caveat: we first tried to recover KLA's category boundaries from spectral content and failed — one number cannot separate ten morphologies. We used the labels the source dataset already provides.

UNSEEN CONTENT BLOCK — 478 IMAGES WITHHELD ENTIRELY

model	trained on it?	PSNR	LPIPS
bicubic ×2	—	20.652	0.458
r2-preal1 (real only)	yes	24.562	0.338
shipped	no	24.302	0.182

+3.65 dB on content never trained on, and 60% better LPIPS than the model that WAS trained on it.

256 → 512 WITH ZERO TRAINING AT THAT SCALE

778 held-out 512×512	PSNR	SSIM	LPIPS
ours	25.392	0.576	0.236
bicubic ×2	20.877	0.410	0.538

+4.52 dB — a larger gain than at 256×256, at a scale never trained on.

HALLUCINATION GUARD

Energy above the input's Nyquist — the band where content is necessarily invented. On their 297 images: ours **0.421** vs bicubic **0.375**. Bicubic cannot fabricate by construction, so it is the floor. **Zero of our outputs cross ratio 1.0, against fourteen of bicubic's.**

The OOD Suite, on the Organisers' Own Test Set

Better than real-only training on every degradation axis of their own images — 3.50 dB ahead under severe noise.

axis	bicubic	SHIPPED	100% real pairs
noise 0.05	23.09	+1.07	+1.01
noise 0.19	19.79	+2.89	+2.66
noise 0.40	15.54	+5.59	+2.09
blur 0.4	19.82	+2.84	+2.64
blur 0.8	19.70	+2.68	+2.57
blur 1.2	19.44	+2.63	+2.56
soft (non-box) kernel	19.74	+2.89	+2.66
MEAN gain		+2.94	+2.31
WORST axis		+1.07	+1.01

BETTER ON EVERY SINGLE AXIS

r2-preal1 — 100% real pairs — has the *best* score of anything we trained on the undamaged test set: 23.866 dB against our 23.632. **It loses on all seven axes the moment the degradation moves.** At noise 0.40 the gap is **3.50 dB**. It is ahead by 0.234 dB when nothing changes, and behind by 3.50 dB when the sensor does. That is the whole argument of this deck, measured end-to-end on the data the organisers gave us.

297 images from the organisers, re-degraded seven ways. docs/tool_output/ood_testset.txt

Result. The shipped model improves over real-only training on every tested degradation axis of the organisers' own images, with the largest advantage under severe noise — **+5.59 dB against +2.09 dB at $\sigma = 0.40$** . It gives up 0.234 dB on the undamaged set and gains 3.50 dB the moment the degradation moves. That asymmetry is the case for this model.

Multiple Runs & Stability

Seed floor 0.005 dB. Identical to five decimals across two GPUs. Our split ranks all eleven models in the same order theirs does.

1 Seed variance

r2-full-s0 23.1038

r2-full-s1 23.1087

0.0049 dB apart — identical configuration, different seed, matched final-5 window at epoch 39. This is the floor everything else is judged against.

2 Final-5 vs best-epoch

Across six 120-epoch runs, best-epoch selection reads **median +0.058 dB** above the final-5 mean — six times the seed floor. Reporting a best epoch would manufacture an effect out of nothing.

3 Cross-machine

The shipped weights score **23.632 / 0.60791 / 0.19288** on an RTX 4090 pod and **23.632 / 0.60791 / 0.19288** on an RTX 4050 laptop — identical to five decimals, different GPU, different machine.

THE STRONGEST STABILITY RESULT WE HAVE — AND IT ARRIVED FROM OUTSIDE

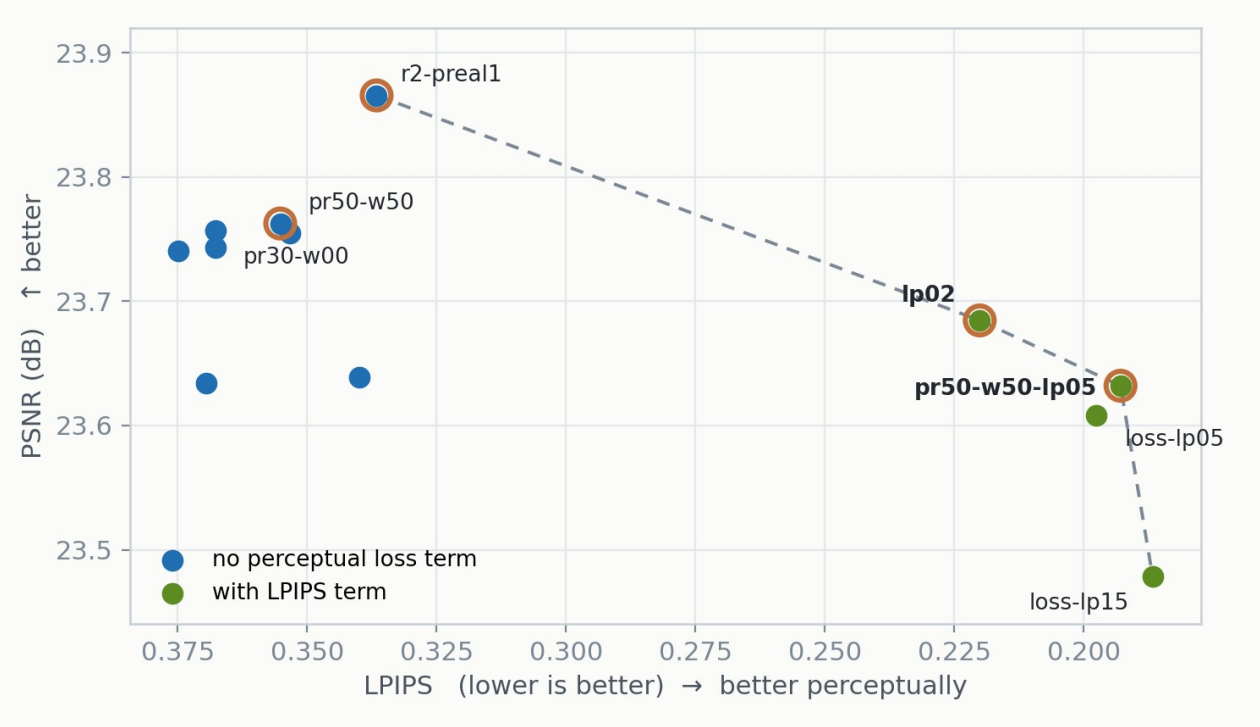
Our entire selection process rests on a 200-image validation split we constructed ourselves. When the organisers released a 297-image paired test set we ranked all eleven models on it and compared against our own ranking:

Spearman ρ PSNR +0.955 SSIM +0.991 LPIPS +1.000

Our split ranked all eleven models in exactly the LPIPS order their data does. The absolute numbers differ — their set is 0.231 dB easier on bicubic — but the ordering, which is what every decision in this deck depends on, is identical. Every comparison in the engineering log was measured on a good ruler, and that is now demonstrated rather than assumed.

Final Model Selection

We did not ship the best-PSNR model. We shipped the one whose weakest axis is strongest — because the downside is 7.6× larger on the perceptual side.



Circled: the efficient frontier candidates. Bicubic ×2 baseline 20.455 / 0.5099 / 0.4655, identical for all eleven.

run	PSNR	SSIM	LPIPS	$\sigma=0.45$
r2-preal1 (100% real)	23.866	0.632	0.337	+1.88
pr50-w50 (best of the wide, no LPIPS)	23.763	0.619	0.355	+6.03
pr30-w00 (no wide family)	23.757	0.613	0.368	+2.36
lp02	23.685	0.607	0.220	+6.00
pr50-w50-lp05	23.632	0.608	0.193	+5.91
loss-lp05	23.608	0.604	0.198	+5.91
loss-lp15	23.479	0.595	0.187	+5.76

THE TRADE, AND WHY IT IS NOT SYMMETRIC

Across the eleven models the worst PSNR gain still keeps **91.4%** of the best. The worst LPIPS reduction keeps only **32.5%**. Ship perceptual and LPIPS is not scored → we lose **6.9%** of achievable PSNR gain. Ship non-perceptual and LPIPS is scored → we lose **52.7%** of achievable LPIPS reduction, plus 4.03 dB of noise headroom. **A 7.6× asymmetry in downside.** Insuring against the large loss costs almost nothing, and that argument holds under every normalisation we tested.

Reproducibility & Submission Readiness

All four required components, each verified by running it. Bit-identical output from a directory holding only those four files.

```
pip install -r requirements.txt
```

```
python run.py <input-dir> <output-dir>
```

THE ENVIRONMENT SPECIFICATION

requirements.txt — the complete pip freeze of the environment that trained the weights, exactly as specified. One line is *added* — the CUDA index URL — because torch==2.13.0+cu126 is not on PyPI and the file would not install without it. No package line altered or removed.

requirements-inference.txt — a convenience, not the spec: the two packages run.py actually imports.

VERIFIED, NOT ASSERTED

tools/package_check.py restores a held-out sample, checks dtype, finiteness, range and shape on every output, then copies *only* run.py, requirements.txt, README.md and models/model.pt into an empty directory and runs it from an unrelated working directory. Prints PASS or a specific FAIL.

THE CONTRACT

In: .npz (H,W) float32, any range.

Out: .npz float32 (2H,2W), same name, finite, in [0,1].

Weights resolve relative to run.py.

No internet, no keys, no downloads, no interaction.

WHAT THE REPOSITORY CARRIES BEYOND THE FOUR REQUIRED FILES

docs/ENGINEERING_LOG.md — every experiment including the failures and eight claims we published and had to withdraw · docs/queue/ — all eleven models scored on both test sets, with the raw CSVs · docs/superseded/ — two result files produced against a different test set, kept with a note explaining why they must not be quoted · docs/bench_results.txt — the timing runs behind three rejected optimisations.

We measured the degradation, trained for variation beyond it,
and validated on unseen content and unseen degradation.

23.63 dB

PSNR · organisers' 297-image test set

+3.18 dB

over bicubic ×2

0.193

LPIPS (bicubic 0.466)

VALIDATED WHERE THE MODEL HAS SEEN LEAST

- +3.65 dB on a content block withheld entirely from training
- +4.52 dB at 512×512 — a scale it was never trained at
- +4.01 dB mean across all ten NFFA morphologies, a different corpus — every one positive
- Better than real-only training on every OOD axis of their test set; 3.50 dB ahead under severe noise

STATED BEFORE ANYONE HAS TO ASK

- The 512 path is verified on NFFA natives, not KLA 512 data — the training set has none
- The wide family's bounds are a judgement about breadth, not a measurement
- We do not know how PSNR, SSIM and LPIPS will be combined — so we optimised all three
- Biological morphology is the weakest class, +2.26 dB against the +4.01 mean

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python run.py <in> <out> · torch + numpy only · 1.37 M params